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ORIGINAL ARTICLE

Glaucoma classification using deep learning ensemble and multimodal image modality

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Abstract

Glaucoma is a group of optic neuropathy disorders related to the degeneration of retinal ganglion cells. It is complex to diagnose for the variety of clinical manifestations and the presence or absence of symptoms. Color Fundus Photographs (CFP) and Optic Coherence Tomography (OCT) are two of the principal imaging techniques used during evaluations by clinicians. An enriched and complementary view of the retina could be reached by using both modalities. In this work we propose a Convolutional Neural Network (CNN) ensemble consisting of two modules (by image type) and then their mixture into a single prediction via fusion techniques (intermediate fusion, late fusion) to classify glaucomatous cases. Our approach achieved acc = 91.88%, prec = 89.71%, recall = 92.84%, F1 = 90.93% and acc = 94.02%, prec = 94.06%, recall = 94.05%, F1 = 94.02% for the ensemble of each respective modality (CFP, OCT). The performance was of acc = 93.60%, prec = 83%, recall = 87.03%, F1 = 84.85% and acc = 86.84%, prec = 87.97%, recall = 86.39%, F1 = 86.61% for two testing databases from mexican and pakistan populations, respectively.

Keywords Glaucoma · Deep learning · Ensemble · Convolutional neural networks · Optical coherence tomography · Color fundus photographs

1 Introduction

Glaucoma is a multifactorial optic neuropathy that affects millions of people worldwide (Chaurasia et al. 2025). It is the most common cause of irreversible blindness worldwide, the second leading cause of blindness after cataracts, and the second most common cause of irreversible moderate or severe vision loss (Bourne et al. 2024). It is asymptomatic at early stages and therefore challenging and urgent to detect early cases because lost vision caused by glaucoma cannot be reversed, but it is preventable.

Tunnel vision is a type of blindness that limits the visual field to only central vision (Shehryar et al. 2020). In this neuropathy, optic nerve fibers progressively degenerate, leading to structural changes in the optic nerve and, therefore, diminish in the visual field. This degeneration means loss of retinal ganglion cells and their axons (Babu et al. 2015; Ophtalmology 2021). Retinal layers are shown in Fig. 1.

Elevated intraocular pressure (IOP) is one of the risk factors that may precede glaucoma. This increased pressure could appear due to defects in the trabecular meshwork, whose function is the drainage of aqueous humor of the eye.

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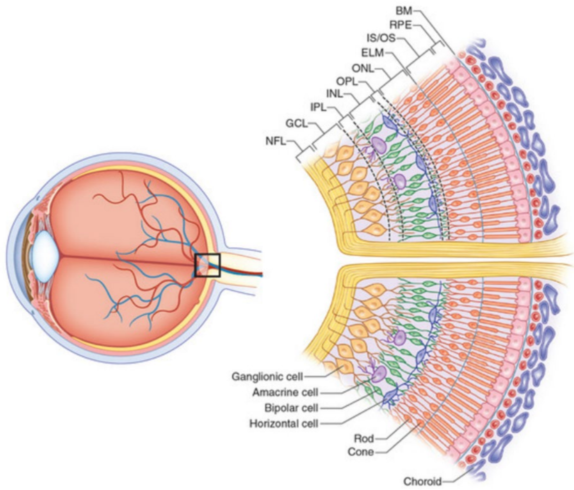


Fig. 1 Retinal layers. In yellow, retinal nerve fiber layers can be viewed. Retrieved from Reference T.R.

Although it being the most frequent, it is not the unique one. Other risk factors can include increased age, increased prevalence in specific races, existing family history, genetics, optic disc hemorrhages, peripapillary atrophy, high myopia, diabetic population, migraine, increased systolic and diastolic blood pressure, for example. It should be understood that glaucoma refers to a large group of disorders with a variety of clinical manifestations. This makes glaucoma a heterogeneous group: it is more common in older people, but there exists in juveniles and infants. There also exist variants with normal IOP (Allingham et al. 2011).

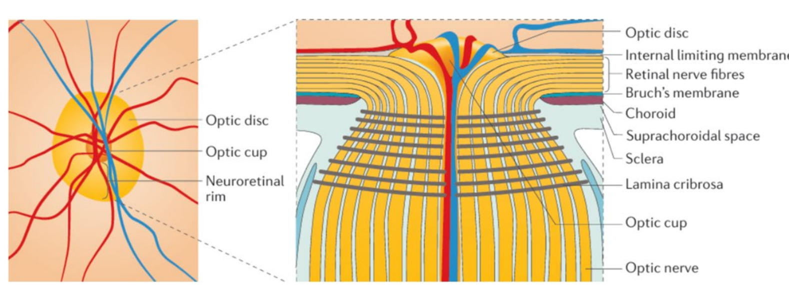
Typically, ophthalmologists use multiple methods to diagnose glaucoma. These include gonioscopy (a study that measures the angle between the iris and the cornea), pachymetry (corneal thickness), tonometry (IOP measurement), and campimetry (visual field assessment) besides imaging modalities like fundus imaging or optical coherence tomography acquired with specialized equipment (Shoukat et al. 2023; Coan et al. 2023). The complementary nature of all these examinations combined with the clinical expertise of the ophthalmologist helps them to decide. However, this is not an easy task. Many variants of glaucoma (Primary Open Angle Glaucoma, Normal Tension Glaucoma, Closed Primary Angle Closure Glaucoma) are painless, and visual field defects are discrete or barely noticeable in the early stages. This may cause a moderate to late diagnosis accompanied by vision loss (Li et al. 2023).

Imaging modalities are vital in the detection of glaucoma. They allow clinicians to visualize and evaluate structural abnormalities in the eye. Such abnormalities often precede the development of visual field loss (Coan et al. 2023). An important structure in glaucoma is the optic nerve head (ONH). It is a region in the retina (the distal portion of the optic nerve) where blood vessels and the axons that emerge from the ganglion retinal cells converge. ONH is often named as optic disc (OD) or optic papilla (Saha et al. 2023). Optic cup (OC) can be understood as a central depression without nerve fibers. Figure 2 shows a frontal and a cross-sectional view of these structures.

Typically, perceptible parameters point to changes in these structures and have been associated with glaucoma severity. Cupping, for example, shows an increase in the size of OC. The cup-to-disc ratio (CDR) parameter measures the difference in size between OD and OC. It has variants that measure just the vertical ratio or the total area proportion (Coan et al. 2023; Takada et al. 2016). Thinning of the Retinal Nerve Fiber Layer (RNFL) is also an indicator of glaucoma (visible in OCT images, slightly in CFP) (K et al. 2024). Other retinal layer measurements, like Bruch's membrane opening (BMO), have recently been shown to be accurate and reliable for glaucoma evaluation (Ran et al. 2021). Peripapillary atrophy (PPA) is the thinning or degeneration of tissues surrounding the ONH. These zones are characterized by evident differences in pigmentation (viewed with CFP). Ganglion cells combined with the inner plexiform layer (GCIPL) have also been found to be a clinically useful glaucoma biomarker, with their thickness showing disease progress (Maetschke et al. 2019). Disc hemorrhages are red or darksome characteristic linear defects perpendicular to OD, and their presence may indicate progression in glaucoma (intensification of IOP).

In contrast, the significance and relations established are tough: PPA is also seen with other conditions such as myopia and aging changes; the size of the OC varies considerably in normal populations, and even though physiological cups tend to be symmetric between the two eyes of a person, the asymmetry itself is not enough to identify glaucoma; RNFL measurement is affected by several factors including

Fig. 2 Optic Disc and Optic Cup. Retrieved from Weinreb et al. (2016)



poor image quality or high myopia, and defects are visible in many neurologic disorders and healthy individuals; disc hemorrhages tend to come and go, reappearing in other locations or sometimes the same. In summary, there is a significant overlap in the ocular features of normal subjects and patients with early glaucoma (Allingham et al. 2011; Ran et al. 2021; Akter et al. 2022).

Changes in this region are visible in either optical coherence tomography (OCT) or color fundus photography (CFP). OCT is a non-invasive imaging modality that uses low coherence interferometry (infrared light of approx. 843 nm) to generate high resolution images of the retina in a cross-sectional view, allowing visualization of deep tissues in real time (Maetschke et al. 2019; Yang et al. 2022). Fundus Photography is a well-established and cost-effective imaging technique (one of the most used) that captures the appearance of the retina (in a two-dimensional way), including OD, OC, and blood vessels (Shyamalee et al. 2024). Figure 3 shows real OCT and CFP images and some normal or glaucomatous examples.

Here, Artificial Intelligence (AI) approaches gain relevance. The diagnosis of glaucoma could be defined as a classification problem in which one or more features (images, clinical parameters) are available and a single diagnostic variable is expected as an output (presence of glaucoma, severity, type) (Ophtalmology 2021). This should not be understood as a specialist substitution but as a support tool. Many proposed systems are based on Computer Aided Diagnostics (CAD). These systems have reached accuracy levels comparable to those of human experts (Muchuchuti and Viriri 2023).

In the clinical setting, specialists typically have access to some devices to take images. Then, they need to send or store the images of a particular patient for analysis and make the acquisition device available to incoming patients. In case the analysis is made in situ, the new patients must wait until the previous case is referred to a particular specialty located either in the same institution or another one. This process is manual, time-consuming, needs the availability of specialists in these institutions and are prone to human mistakes

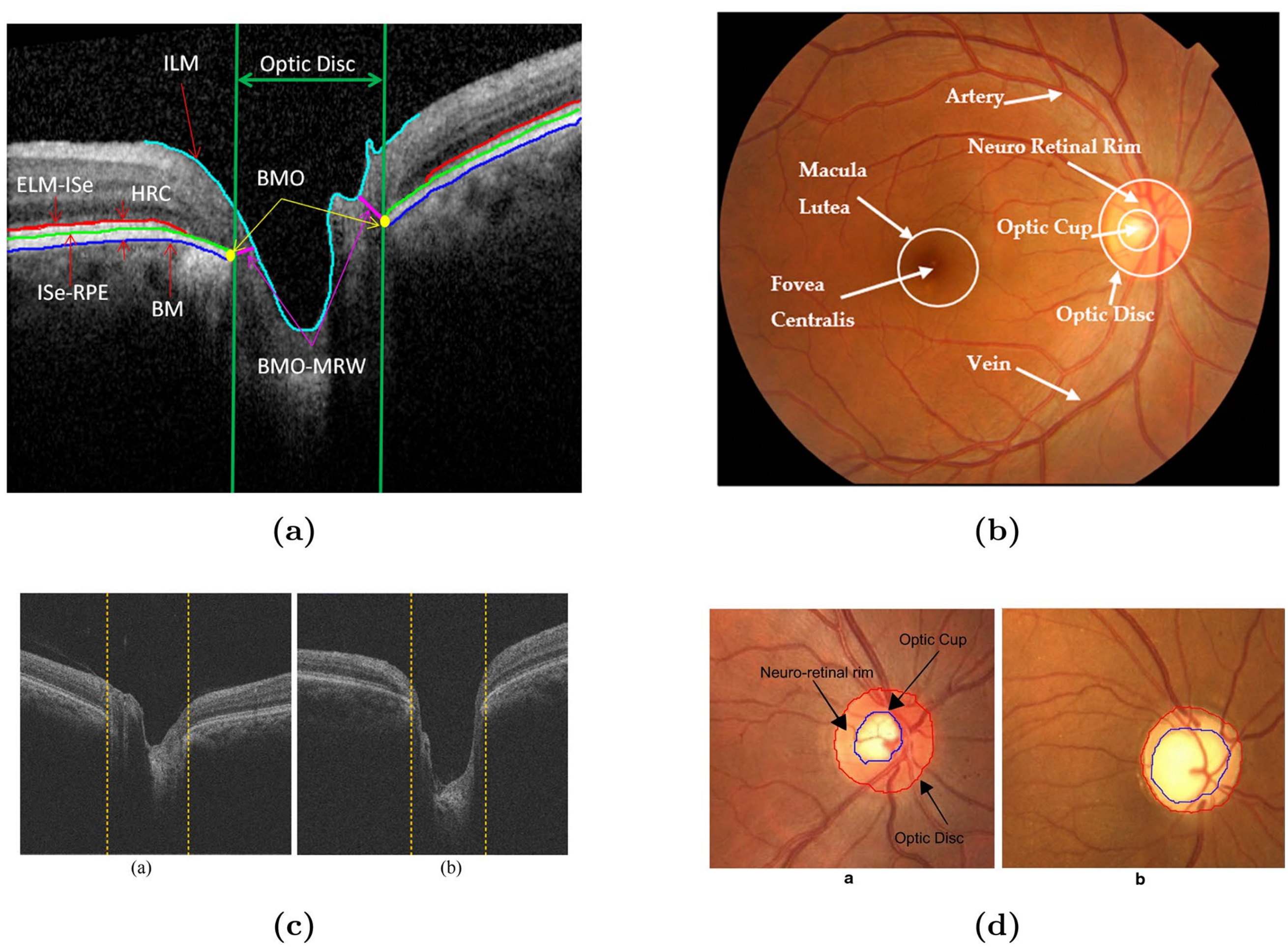


Fig. 3 Typical Structure of **a** OCT and **b** CFP images. Examples of **c** Normal and enlarged cup in OCT and **d** Normal and Enlarged cup in CFP. Elaborated from Zedan et al. (2023); Hussain et al. (2015); Yu et al. (2018); Diaz-Pinto et al. (2019)

due to fatigue or less experienced clinicians. The development of a support tool for an automatic classification could relief the workload in this context, allowing faster, efficient and cost-effective screening.

Many of the systems developed so far are based on Deep Learning (DL), a Machine Learning (ML) subarea, which uses neural networks to extract abstract but relevant characteristics (named features in this field), allowing the distinction of the presence of neuropathy. Particularly, models are mainly dependent on CFP due to the acceptability of the imaging technique, their cost-effectiveness, and their large-scale screening and diagnosis of glaucoma (K et al. 2024; AlRyalat 2021). As one of the main drawbacks, the DL algorithms focused on CFP alone rely on the detection of structural changes in glaucoma from the top view of the ONH and the retina, without considering the inner layers or entire structures (Ran et al. 2021).

Currently, there exist medical devices that acquire both CFP and OCT on the same machine. Therefore, the development of AI systems which incorporate both types of images in their process of classification, is plausible. Also, the advances in computer processing power allow this kind of approach that otherwise may not be possible due to memory constraints. With that in mind, a multi-image solution has the potential to achieve an enhancement in detection by combining these image pairs. Doing that imitates human reasoning (by comparing information), processes huge amounts of data and still achieve high accuracy in the distinction of disease.

The motivation of this work is the development of a system capable of integrating information from different imaging sources, particularly OCT and CFP images, and perform the automatic classification of cases as glaucoma or not, using a Deep Learning approach. The DL approach is selected because it allows us to analyze large amounts of information without the need for hand-crafted features and still has great accuracy performance. Early diagnosis is critical, because the blindness caused by the disease is irreversible, and an automated system also alleviates the manual and intensive labor of clinicians. We also expect the system to be more robust (in terms of overall performance) by executing this mixture of modalities, which could be benefic compared to rely on a modality alone, as this kind of models focus on different views of the same anatomical structure.

In this study we propose a model for the classification of glaucoma based on convolutional neural networks. We selected several pretrained models as our baseline and trained fully connected layers on top of them, then we performed fine tuning on each of the models. Later, we combined the model for each type of image separately with different fusion methods for building an ensemble per modality. Then we combined the different types of predictions by each image type. Finally, we validated our approach in two external databases and validated that our approach achieves better performance than a single model or modality.

The contributions of this work are the following:

1. The development of an initial Deep Learning model (CNN based) for each image modality alone, as a baseline.
2. The exploration of diverse ensemble techniques, and the selection of the more adequate one for the fusion.
3. The mixture of different modalities (CFP, OCT modules) in an ensemble.
4. The incorporation of Mathews Correlation Coefficient to our metrics, generally not reported in Deep Learning model performance.
5. The validation of our approach on two external databases without retraining the models for each use case.

This paper is ordered as follows. Section 1 comprises introduction and literature overview. Section 2 presents the datasets used for training the models, the preprocessing applied, the selected DL models and the techniques explored for performing the ensemble in our approach. In Sect. 3 the results of the individual CNN and the ensemble techniques are presented. Section 4 discusses and analyses these presented results and finally, Sect. 5 concludes the paper.

1.1 Related work

Studies have already addressed glaucoma detection using two or more modalities to enrich or complement their results (Shehryar et al. 2020; Babu et al. 2015). Modalities refer not only to images (such as OCT or CFP). Clinical history or visual field studies were also integrated in some approaches (Ophtalmology 2021) and some unimodal studies report the need to acquire multi-modal data to combine predictions and perspectives to develop new models (Shoukat et al. 2023; Zhang et al. 2023). Convolutional Neural Networks (CNN) are a popular technique in the DL field because they have the capacity to learn relevant features and patterns directly from images, preserving the spatial structure of those images. They have been shown to be beneficial in medical imaging classification (Pascal et al. 2022).

For example, Christopher et al. (2018) used three different CNN architectures (VGG16, Inception, ResNet50) to classify CFP images as glaucomatous optic neuropathy (GON) or healthy from a racially and ethnically diverse group. It achieved an AUC of 0.89–0.91 in GON detection and identified any, mild or moderate to severe GON with AUC (0.91–0.97), Sensitivity (0.84–0.92), and Specificity

(0.83−0.93) in their three-class approach. The errors were due to a large CDR (Healthy predicted as GON).

The research in Li et al. (2018) also with CFP images, also classified the images as GON or healthy. Twenty-one trained ophthalmologists classified the images as ground truth using the Inception V3 architecture. They achieved AUC = 0.986, Accuracy = 92.9%, Sensitivity = 95.6%, Specificity = 92%. This study was conducted using 39,745 images, and the most frequent reason for misclassification was physiologically large cupping.

The approaches of Saha et al. (2023) and Shyamalee et al. (2024) explored interesting systems. The first used a two-stage approach: automatic region of interest (ROI) extraction from CFP (centering it at ONH) using the simplified YOLO Net architecture and posterior classification with the Mobile Net architecture. Their findings include comparing this lighter CNN to more complex and deeper architectures (ResNet50, InceptionV3, InceptionResNetV2, among others), achieving similar performance with the least memory requirements. They also pointed out that, in an experiment to classify the whole image, there existed an accuracy drop from 2–3% in all models. The second study also used a two-step approach: 1) CFP ROI cropping with segmentation of OC and OD using a modified ResNet50 architecture, and 2) classification as Healthy or Glaucoma using a modified Inception V3 architecture. Their results include accuracy = 98.97%, precision = 99.41%, sensitivity = 99.42%, specificity = 95.99% and AUC = 0.99 for one publicly available dataset named RIM-ONE.

On the OCT image side, Akter et al. (2022) extracted the ONH cup area from B-Scans (segmentation) and trained a custom CNN of 24 layers for glaucoma prediction. They compared their results with popular architectures for OCT-based glaucoma classification (ResNet18, VGG16), achieving similar results (AUC = 0.99, accuracy = 98.6%, sensitivity = 99.4%, specificity = 97.8%, precision = 97.8%) and demonstrated that the ONH cup area of the B-scans could be an imaging feature to be added in glaucoma diagnosis.

Maetschke et al. (2019) used a 3D CNN to classify eyes as healthy or glaucomatous from raw, unsegmented OCT volumes. They compared their approach to other ML techniques (SVM, Logistic Regression, Random Forest, etc) and found higher AUC when using CNN (0.92 in contrast to 0.89 in the best ML model). Their 3D CNN identified neuroretinal rim, optic disc cupping and surrounding areas as the more important ones for the classification.

Moving to the multi-modal image-based approaches, the work done by Babu et al. (2015) compared a backpropagation neural network (BPN) with a Support Vector Machine (SVM) as the classifier. They first extracted features for each image type (CDR and ISNT ratio for CFP, cup depth, and retinal thickness for OCT) and then used the combined features to perform classification. They reached a sensitivity = 98%, specificity = 93.33%, and accuracy = 96.92% for the SVM classifier. However, their database was limited. One hundred and fifty dual images were used (50 were normal and 100 were glaucomatous), and part of these images were used to train the classifiers (n = 85).

The research by Shehryar et al. (2020) integrates data by modularizing each modality first. Then, each module extracts either structural (cup shape and rim-to-disc ratio in CFP, CDR in both CFP and OCT) or textural and intensity-based features (115 textures for CFP, no textural features for OCT). Then, classification is performed based on SVM, and the overall system was validated in a database comprising 50 dual images, achieving accuracy = 96%, specificity = 91%, and sensitivity = 100%. Final classification was decided in the case that the fundus and OCT modules agreed (glaucomatous, normal) or suspicious if the modules disagreed.

Those last two works did not use CNN as their classifier, and followed an intricate process to extract the features of each imaging modality, which may affect their application to vast amounts of data. Additionally, the number of dual images for testing the developed system was small and was the only data used to evaluate their results.

The works consulted despite being promising have limitations: Many of them do not validate their approach on an external dataset (Babu et al. 2015; Saha et al. 2023; Li et al. 2018), some retrain the model for each database used instead of mixing or holding out one or more databases and report the best dataset performance as their main results (Shoukat et al. 2023; Shyamalee et al. 2024) and others rely on just one database during training and testing (Maetschke et al. 2019) or equipment for data acquisition (Christopher et al. 2018) limiting the evaluation of system generalization.

Although the consulted studies are valuable and achieve good performance, their techniques can be further discussed. For example, Christopher et al. (2018) achieved better performance when using a transfer learning approach against a training from scratch approach on the same network architecture. This allowed us to avoid trying to train a complex architecture from scratch. However, they did not explore the fine tuning of their transfer learning methods, even when this could be benefic by specializing the network in distinguishing medical images. The study of Li et al. (2018) highlighted the usage of regularization techniques to prevent or limit overfitting (dropout and a large dataset). However, their CFP images were only from Chinese hospitals, they only used the Inception network as their classifier, they did not use a visualization or explainability technique and performed the classification on the whole image.

The proposal of Saha et al. (2023) is one from that we took their findings of cropping the CFP as a starting point. They also used transfer learning and incorporated fine tuning, but they do not explicitly say from which specific layer they used it in their final models. Their learning rates are also specified in a range, but no final learning rate, scheduler, or optimizer are detailed, just referred as the optimal setup. Also, there is no evidence in this work of the usage of a visualization or explainability technique. Even with these drawbacks, it is remarkable the performance of their model, with significantly less memory than state-of-the-art-ones.

In Shyamalee et al. (2024) they incorporated explainability in their classification approach. In particular they used the GradCam and GradCam++ techniques for this purpose. By observing the examples they provide of the heatmaps generated, we opted for use GradCam, as it is simpler in its calculations and the heatmaps produced are similar, and differ in border and contours in most cases. If the developed model is precise enough, the main objective of these visualizations is to generate trustworthiness by highlighting the approximate zones used for decision, which is an objective achieved by GradCam in many cases.

The investigation of Akter et al. (2022) stands out by demonstrating the outstanding performance of a DL approach against a traditional ML one. In its investigation, their specificity, accuracy and AUC were better in the DL model (simple multilayer perceptron) designed with selected features. Then, the comparison of pretrained CNN against a custom CNN was executed. In this comparison, it is not specified if they use fine tuning on the pretrained models (which we assume not). This is important, because their proposed architecture seems to be better, compared with these more complex architectures. The performance could be superior in their research due to the quantity of available data. For the training of their CNN, they report 360 balanced images (binary masks of OCT slices). That represents the 80% data, which corresponds to 6 slices of the OCT volumes. The testing database is reported as 20% of the total of images available. We take these results with caution, because the small size could exhibit better performance in the network.

The system proposed by Maetschke et al. (2019) performed at volume level their classification. Here is where early stopping as technique for saving the best weights is mentioned. It is to note that this 3D-CNN was also customized with barely five convolutional layers. In this study, GradCam is again used, which reinforces the popularity and suitability of this visualization technique. The study reported the AUC in the validation and test phase alone, with no additional metrics. The authors were aware that a deeper, complex or different architecture could have reached higher performance, but their limited GPU memory limited the demonstration.

The investigation of Babu et al. (2015) despite being less recent, performed the OC and OD calculation by using a technique called Spatially Weighted Fuzzy C Mean Clustering (SWFCM), and for the detection, the blood vessels were blurred. Using that technique, they achieved less segmentation errors compared with a normal Fuzzy K-Means (by considering the neighboring pixels). This method was not dependent on the device. They also calculated the ISNT areas as a feature, where the ranges measured in the different classes can be used for distinguishing images. On the OCT side, they used cup depth as a feature. CDR was also incorporated. They were calculated by applying wavelet transform and multilevel thresholding for retinal choroid and retinal vitreal boundaries. Retinal thickness was measured too, but for the calculation they needed the resolution factor of the camera that captured the images. As a drawback, it has been found that ISNT rule is valid only for one third of patients, approximately, while other variants are valid in more than 70% of patients (Poon et al. 2017; Corredor-Arroyo et al. 2022) The results of this study were obtained on the same database where they trained the SVM and BPN (no external validation), and report the need of higher resolution fundus images for achieving an accuracy improvement. No details of image resolution, acquisition device or population were provided.

The work of Shehryar et al. (2020) performed the disc and cup segmentation and size measurements using an enhanced image, binarized it, cleaned noise and then used the Canny Edge detector and regionprops for the diameter calculation. The problem with this procedure was that the original shapes of OC and OD were selected empirically. In fact, the selected thresholds were applied for different color channels (red for disc segmentation, blue for cup segmentation). This is prone to errors, in case the images differ in illumination, noise level, device type, for example. In contrast to that procedure, the study performs a cup shape analysis, where irregular cup contours suggest a healthier eye against an irregular one. This is based on the evidence that blood vessels patterns are more complex, irregular and dense, for example, in healthy eyes. This second feature is computed using Hausdorff's fractal dimension, and we consider it to be adequate to fulfill the classification purpose. Other features like the previous one are designed, until they reach around 100. The validation in their method is valuable, because they did it over a local database and two public databases. On the OCT side, they extracted the internal limiting membrane (binarization of image, surface extraction, noise removal, missing points filling) and the retinal pigment epithelium layer. Those layers were segmented to perform OC and OD diameter calculations. That process of segmentation was not perfect. In fact, in their approach they had to define a CDR = 0.2 manually, in case cup edges were not detected, which may influence the classification accuracy. They did not say if they designed their OCT module for the same or different images that they used for the validation of the overall system. For the images they provide as examples during the explanations, we assume the images used are from the same device, population and type. This could limit the generalization or system validation on different devices or populations. It also may have influenced the greater performance reported in the OCT module. Also, that OCT module had only one feature (CDR calculation) in contrast with the more than a hundred present in their CFP module.

In our investigation, we did not find researches that use CNNs as their models for a multi-modal approach using enough data for each modality. Based on the evidence that glaucoma classification using 2D-CNN either using CFP or OCT works accurately, the ease of integrating large volume data on DL approaches, the automatic extraction of features by using feature maps in convolutional networks, learning progressive complex hierarchical characteristics and the need to develop ensemble technologies instead of a single model for robustness, we propose a multi-modal deep learning (CNN) ensemble approach based on CFP and OCT images, validated in two separate external databases of dual images, and explore various fusion methods.

2 Materials and methods

Modularity was also adopted in our approach as multimodal works do. We developed two modules, each one for one image modality, with different CNN architectures. We will begin by presenting the CFP module, data used for training the CNNs, preprocessing done for these images, architectures, and hyperparameters used. Then, we will present the same information for the OCT module, and finally, we will discuss the fusion methods explored for building the ensemble.

Table 1 Databases used for training the CFP module

Dataset	Country of origin	Glaucomatous	Healthy	Total
ACRIMA (Diaz-Pinto et al. 2019)	Spain	396	309	705
RIM-ONE-DL (Fumero et al. 2020)	Spain	172	313	485
ORIGA (Zhang et al. 2010)	Singapore	168	482	650
LAG (Li et al. 2020)	China	1711	3143	4854
Sjchoi86-HRF (yiweichen: Sjchoi86-HRF database. https://github.com/yiweichen04/retinadataset)	Unknown	99	294	393
BEH (Islam et al. 2022)	Bangladesh	171	463	634
				7721

2.1 CFP module

This module is composed of 3 different CNNs trained on publicly available data. When training DL algorithms, a large quantity of data is needed to avoid overfitting. We mixed different publicly available databases for the purpose of training on enough examples and incorporating a diversity of images (acquisition, demographic diverse population). Table 1 summarizes the public databases used, the quantity of original images, and information on whether the examples are glaucomatous or not.

We followed the findings of previous works that indicate a performance drop occurs when predicting in the whole image rather than in a cropped version centered in ONH, differences in acquisition devices may amplify differences in images or indicate that cropping fundus images and training CNN proved to be satisfactory in detecting glaucoma (Saha et al. 2023; Islam et al. 2022; Asaoka et al. 2019). Thus, we cropped a region of interest (ROI) around the optic disc. To do so, a kernel was scrolled through each image to calculate the average intensity of pixels on each image portion, and then all the intensities were compared to determine the region with the major intensity. That region was expected to contain the OD. With that value determined, the central coordinates in the image patch were recovered, and a window size was defined for the final ROI, as approximately 20% of the original image. ROI extraction was performed in the databases that did not provide the images already cropped. Examples for correct and incorrect OD detection are shown in Fig. 4. Most images were successfully cropped. For the ones that were not (borders too bright), manual crops were applied. Image quantity of undetected ONH was variable between databases, but it remained under 10% in all databases.

Before ROI selection, image preprocessing was applied. This was done to enhance the image aspect and the visible structures. We followed the algorithm proposed by Alwazzan et al. (2021) consisting in applying a Wiener filter followed by Contrast Limited Adaptive Histogram Equalization (CLAHE) technique to the green channel. CLAHE technique was used because it is an improvement compared with adaptive histogram equalization (AHE), a more classical algorithm where background noise was amplified even when interest structures were improved. CLAHE corrects this by limiting the quantity of pixels associated to each intensity level, redistributing them to other intensity

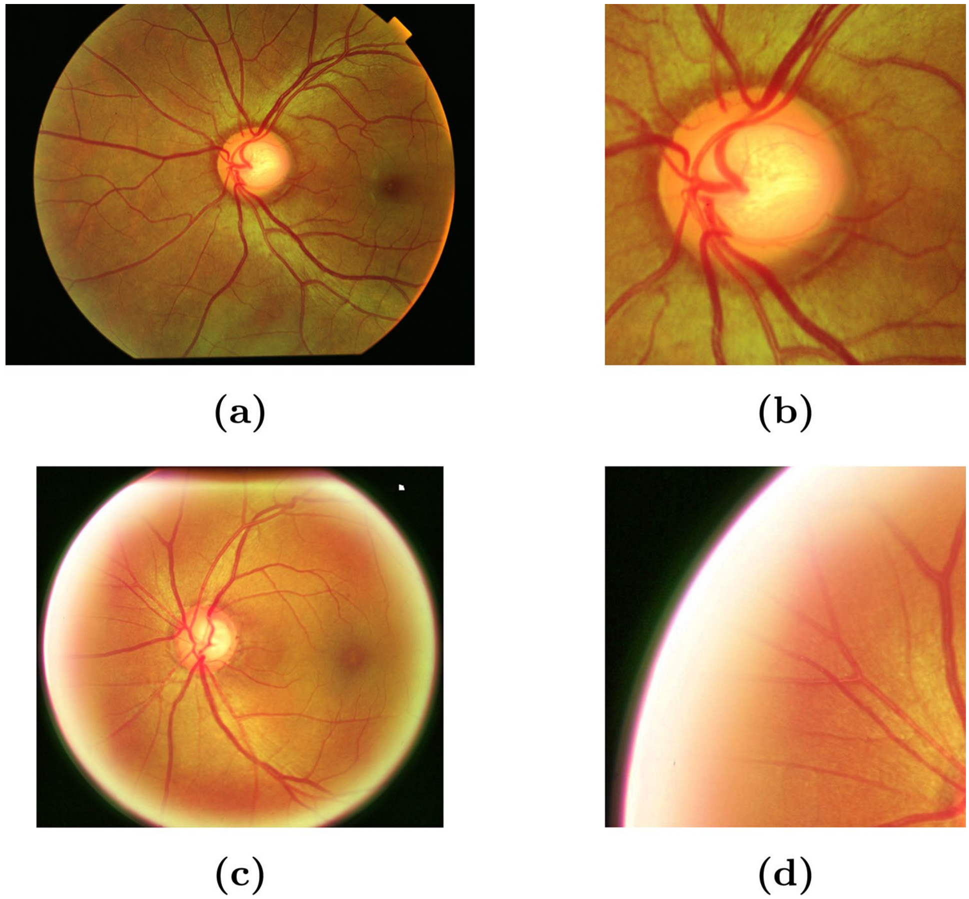


Fig. 4 Images **a** Original, **b** Correctly cropped, **c** Original, **d** Incorrectly cropped. Images are from BEH and ORIGA databases

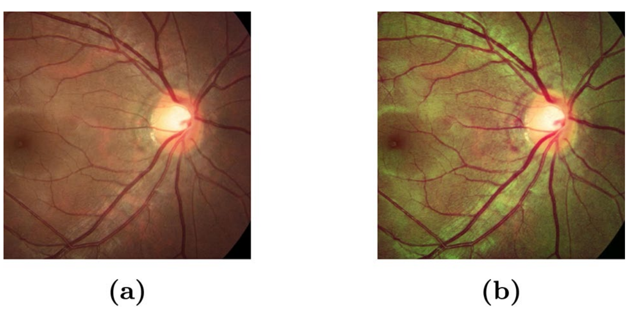


Fig. 5 Images **a** Original, **b** Preprocessed. Images from the LAG database

levels, allowing a smoother transition (Zuiderveld 1994). Combining the filtering and CLAHE to one color channel demonstrated the enhancement capacity and the mitigation of artificial borders (or noise) produced by the contrasting technique. The enhancement application to the green channel responds better than other channels. This may be due to the image's histogram: In many CFP the green channel dominates the color distribution, covering a broader spectrum of intensities. An example of this preprocessing is presented in Fig. 5.

After the preprocessing, ROI extraction and mixture from public databases, we performed the CNN training. We used 70% of the data for training and the remaining 30% for validation and internal testing (15% each). Data augmentation techniques were applied. Horizontal flips, random rotation (0–20 degrees), contrast, sharpness, saturation and zoom changes were performed. These augmentations were implemented with the TensorFlow Sequential API. The augmentations described were selected and applied sequentially. On each epoch on the training CNN, the example for a specific image was different, within the range specified. Those ranges were selected because the alterations were different enough to not alter the original images in a way structures were not distinguishable, or the information was lost. For evading the alteration of the Optic Disc shape, for example, we avoided shearing transformations.

We also noticed that after our preparation, our classes were imbalanced (40% for the glaucomatous aspect, 60% for the healthy aspect). To overcome this problem, classes were weighted during model training. The pretrained networks selected for this module were ResNet50, Inception V3, and DenseNet169. The first has proven highly effective for image classification, object detection, and feature extraction. The skip connections in this architecture solve the vanishing gradient problem, allowing deeper networks without degradation in performance (Shoukat et al. 2023).

The second one is also popular in classifying images with a glaucomatous aspect. It stands out due to its inception modules, "spatial factorization" modules, capable of obtaining general and local features that are mixed by a concatenation filter, and fed to the next module as a single output (Alayon et al. 2023). The third one is characterized by its improved flow of information through the network (and alleviation of the vanishing gradient problem) using all the feature maps of preceding layers as inputs for the following ones. This allows feature reuse and propagation, causing less redundancy and fewer parameters compared to other CNNs (Huang et al. 2017).

For the training, the versions available in Keras (Chollet XXXX). were used. They were initialized with pretrained ImageNet (Russakovsky et al. 2015) available weights, and training was performed in two stages. In the first stage, a global average pooling (GAP) layer was added for reducing dimensionality, followed by several fully connected layers (three of them with 512, 256 and 128 units) each one with their respective techniques for regularizing (dropout of 0.4, L1= 1e-5, L2=5e-4) and with activation ReLU (Chollet XXXX). The usage of penalty terms was selected because it allowed us to both maintain small neuron weights (L2) and get rid of the less important features (L1). Each penalizing term incorporates a factor into the loss function by the calculation of the sum of the absolute weights (L1) or the sum of the square of the weights (L2). Our dropout rate was selected as a compromise between less dependence over a specific set of neurons and the inability of our model to learn. We saw through experimentation that a very high dropout caused the inability to learn effectively and caused saturation in learning, and a smaller rate increased the overfitting issue. This technique is one of the more useful ones, capable of reducing overfitting, besides being simple and intuitive (some neurons in the layers are randomly ignored, forcing the model to learn from all the weights and not only a specific set of neurons). The GAP layer was selected because it is an alternative to the flatten layer, preferred for producing less parameters (acting as a regularizer), being less dependent on dropout and their robustness to spatial translations of the input (Lin et al. 2013). In this initial stage, training was performed for 30 epochs with an SGD optimizer (Ruder 2017) and a learning rate of 0.001. Using SGD, we expected to obtain a reasonable baseline for further improvement. After this stage, fine-tuning for the entire CNN was performed. The second stage was trained for 80 additional epochs, with an ADAM optimizer (Kingma and Ba 2017), a lower learning rate (1e-4), and caution against overfitting. The initial phase of training was to ensure the added layers in top of the base model, saturated its learning capacity and to take that point as a starting one for fine tuning. ADAM was selected in this second stage for its suitability for optimization problems with a large quantity of parameters and their reported good performance. The lower learning rate during the second stage was chosen to not destroy or unlearn the previous weights by the pretrained model. It is a recommended workflow. Higher learning rates were avoided because the training became unstable and noisy, and lower ones as a starting point begun to learn too slow. We used early stopping and reduced the learning rate if the model stopped improving. The complete tuning of our models was because the task was too different and difficult to detect than ImageNet cases, and other authors have highlighted that deep tuning is a reasonable option in terms of performance (Diaz-Pinto et al. 2019; Tajbakhsh et al. 2016) compared to a training from scratch approach. Moreover, a study that fine-tuned different convolutional blocks of CNN pretrained on natural images, applied to medical images (histopathology dataset), showed that for deep networks, fine tuning the entire network can produce better results than just fine tuning the top or last layers (Kandel and Castelli 2020).

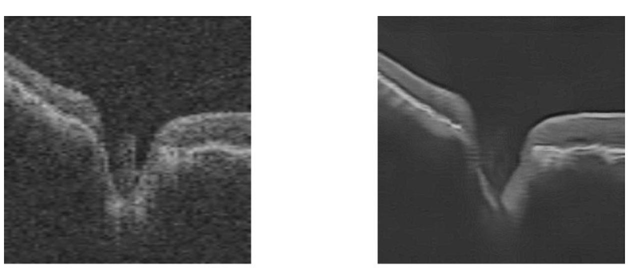
2.2 OCT module

We also trained three different CNNs on publicly available data for this module. OCT images can typically be centered on the macula or ONH. As we focused on ONH as the primary zone of interest, according to our search in previous works, we aimed for OCT images centered in ONH. This limited the open research data. Most open access databases have more representation of conditions such as age-related macular degeneration (AMD), central serous retinopathy (CSR), diabetic retinopathy (DR) and diabetic macular edema (DME) with a bias to more common (or clearer diagnostic criteria) diseases (Rozhyna et al. 2024). The only available database with enough data to be used in our approach was proposed by Maetschke et al. (2019). This database consisted of 1110 scans (volumetric data) centered on ONH acquired from 624 patients on a Cirrus SD-OCT Scanner (Zeiss, Dublin, CA, USA). Patients diagnosed with POAG corresponded to 847 scans, and 263 were diagnosed as healthy, and volumes were 64x64x128.

This work, and several others, have been developed using a 3D CNN approach. Consequently, the labels are determined for the entire volume, not a single extracted image. We intended to classify based on a single central image, so we extended the label to each image. The middle part of B-scans in normal and glaucomatous cases has a relatively higher response and can characterize structural changes or defects. This validation has been done in 3D CNN with selected central scans, which a model trained on selected central scans can yield improvements around 1 to 5% (Wang et al. 2020). For our purposes, we extracted the first 3 to 9 central scans in each volume and labeled them as glaucomatous or not. Then we obtained 2540 images for the POAG case, and 2367 for the healthy aspect, with no class imbalance issues.

The OCT images were not cropped, since they were already centered in the ONH. The preprocessing step was just noise removal. These images are prone to the presence of speckle noise, which may impede clear visualization of the layer structure or cause the system to learn incorrect features. A non-local means (NLM) algorithm was used to denoise the images. It is a non-local averaging algorithm of all pixels in an image. It compares small patches centered on other pixels (in the defined neighborhood) and performs a weighted average only if the patch compared is like the patch of interest, restoring image quality with less blurring and texture alteration of the original image, if used properly (Buades et al. 2005). This kind of algorithm has been used to preserve edges, enhance signal to noise ratio and still being computationally efficient. It has also been used in the field of medical images to denoise MRI images (Li et al. 2024). We used the version provided by OpenCV to perform the NLM filtering. To avoid excessive blurring (generally, OCT images present aggressive noise), an edge enhancement filter was applied after using this technique. An example of the denoising is shown in Fig. 6.

CNN training was performed similarly to the CFP module: 70% of the data was used for training, and 15% for either validation or internal testing. Data augmentation techniques were also applied: horizontal flips, image rotation (0–60 degrees), alterations in contrast, sharpness, saturation, zoom, and translations in 10% pixels in vertical and horizontal directions. For this module, the selected architectures were MobileNetV3 (Howard et al. 2019), Xception (Chollet 2017), and ResNet50 (He et al. 2016). MobileNet architectures are more compact and efficient, typically used in devices with limited resources. They also have been developed to reduce the latency. They are based in depthwise separable convolutions; which allow them to have less parameters; linear bottlenecks, which prevents information loss and comprised representations; and squeeze and excite blocks, which allow compression of feature maps per channel and weight of canal importance. and Xception is a variant of the Inception architecture which uses separable convolutions. That refers to convolutions applied to each input channel individually, followed by a linear combination stage that fuses those features. This allows better information flow through the network, separation and selection of channel importance and better performance (Alayon et al. 2023).



(a) (b)

Fig. 6 Images **a** Original, **b** Denoised using NLM algorithm. Images are extracted from volumetric OCT data from Maetschke et al. (2019)

Training in this module was performed analogously to the CFP module. A global average pooling (GAP) layer was added, followed by their fully connected layers (512, 256, 128 units) and regularizing techniques (dropout of 0.4, L2 of 5e-4). 30 epochs and an SGD optimizer for the first stage, and 80 additional epochs with the ADAM optimizer were also maintained.

2.3 Ensembling techniques

After training our base models (pretrained CNN) for each module, we explored different techniques for the final prediction. The objective of an ensemble is to enhance the predictive power of individual models. The capability of one model may be limited, so the complementarity of multiple classifiers is key. The main idea in an ensemble is to introduce diversity among individual classifiers (Nanni et al. 2023) even if they are homogeneous (same type of classifiers). Among the approximations, we selected late fusion (averaging methods, voting techniques) and feature-level (intermediate) fusion (stacking, multilayer perceptron) for their comparison and selection of the best suited for our application. Those techniques were selected because of their effectiveness and intuition about how they are built.

In classical Machine Learning, the most used and popular techniques for building an ensemble are bagging, stacking and boosting. Bagging consists in training different models independently, on different subsets of data (Nanni et al. 2023). It is a parallel approach. It is expected that the combination of all these classifiers produces different solutions in the classifier owing to the small data changes during training. One advantage of the bagging technique (also known as bootstrapping, or bootstrap aggregating) is variance reduction, which causes this kind of ensembles to generalize better. Problems with this technique are the computational cost (the number of iterations defines the quantity of models trained, and if the model is complex, increasing the quantity may not be feasible) and interpretability loss (Mohammed and Kora 2023). Boosting is another ensemble technique that can be understood as a sequential model, or additive model. On it, base learners (or weak learners) are trained sequentially. Each subsequent model tries to modify or correct the mistakes made by the previous ones. Advantages of the boosting approach include the assignation of higher weights on misclassified or difficult cases. It is also easy to interpret, but a problem is that the next classifiers must correct the errors in previous models. It is prone to overfitting and computationally costly. Stacking needs a two-step approach. First, weak learners are trained, and then a meta-model learns how to combine effectively those base learners' predictions. This technique allows to capture complex relationships between the base models. Typically, a stacking is better than the prediction of a weak model alone (Mohammed and Kora 2023; Auzine et al. 2024). Stacking is also referred as stacked-generalization or stacked ensembles. They are often built by merging heterogeneous weak learners, but it is possible to build an ensemble from homogeneous models. An important thing to note in stacking is that using the same data to train weak learners and meta-model, increase the risk of overfitting (Zhou 2012). Disadvantages of the stacking technique include the risk of overfitting, a need for a large quantity of data, and an increased training time. This last one is because a stacking could consist of several meta-models, complexifying the ensemble (Mohammed and Kora 2023).

On the other hand, there also exist combination methods, here referred as late fusion ensemble methods, since their objective is to combine previous existing models to achieve a strong generalization ability. This is a simpler yet beneficial approach. Reasons behind the selection of these methods is that individual models could get stuck in local minimums, and the combination of different models could serve as different starting points. This implies a reduction in choosing the wrong local minimum. Also, combination methods are a straightforward approach. In fact, methods like bagging or boosting use simple or weighted averaging to combine their generated models. In general, combination reduces the variance and bias (Zhou 2012).

An ensemble could be influenced by different characteristics. They are dependent on the weak or base models trained, their framework (either parallel or sequential), the fusion methods they use to combine their outputs, and the heterogeneity of those base models. Stacking and bagging are parallel methods, and boosting is a sequential one. Heterogeneity refers to the nature of the base classifiers. Same type (like our base CNN) are named homogeneous models while heterogeneous combine base models different in structure. CNN are selected to train our base models as said before due to their large capacity to process large amounts of data along with the preservation of spatial structure of the input data in this application. Deep Learning ensembles have been addressed previously. It is well known that building an ensemble on DL models is more difficult than ML models. One of these reasons is the millions of parameters these models have, and the time needed to train just one model. Another is the difficulty in reaching diversity in the homogeneous base models. Interestingly, when using a CNN based approach for building an ensemble, for medical image classification, voting methods (or late fusion ensemble) are the most used ones, followed by stacking and boosting (Mohammed and Kora 2023). This could be because of the simplicity of the approach, scarcity of medical image databases or large datasets and decent performance.

To ensure diversity among the classifiers, there could be followed different techniques. Diverse preprocessing methods could be applied. The choosing of different architectures, variations in kernel sizes and depth in models could also be beneficial. Using a cyclical learning rate, and each time the model converges during the different cycles, and storing those weight parameters as a separate model, for combining them has also proven to be possible (this process is called SnapShot) (Harangi et al. 2023). The study of Cho et al. (2021) trained 56 CNN for classifying CFP images, using only two base architectures, and for an increased diversity varied their last FC layers from one to three, and the filters applied during preprocessing, before training the model. Our approach to ensure that diversity was to choose those different model architectures and the randomness within a range during the data augmentation on the training database for each trained model. The two image sources also allowed us to limit the quantity of base models used, because the same problem was tackled from two different points of view.

For the ensemble, we selected all the existing voting methods for the reasons mentioned before, their ease of use and low computational expense to combine the base models. The stacking technique and the multilayer perceptron ensemble were selected because those techniques cover the parallel side of the classical ensembling approaches. The small multi-layer perceptron is selected because CNN features mixture it is a simple but effective approach in the combination of different modalities (Li et al. 2022), can handle the differences in modalities and in general, the performance is improved compared with a single modality scenario.

2.3.1 Late fusion ensemble

Majority Voting (MV), averaging (AV), and Metric Based Weighted Voting (MBWV), a variation of Accuracy Based Weighted Voting, were the voting methods used for the late fusion ensemble. MV takes the final prediction made by each model, and the final decision is the class with the most votes from the base models. Mathematically, it can be described according to Eq. 1.

$$N(V_j) = \sum_{i=1}^{N} V_{ij}$$
$$O = Max\{N(V_1), N(V_2), ..., N(V_j)\} \tag{1}$$

Where N(V_j) is the number of votes for the j_{th} class, V_{ij} is the vote for the j_{th} class of the i_{th} classifier, and O is the predicted class after all votes (Rehman et al. 2021). This is the most commonly used technique in voting-based decision fusion methods (Velpula and Sharma 2023). AV is almost equal to MV. The difference is that all the classifications are averaged for each class and the final decision corresponds to the higher averaged class between models. The differences with respect to MV can be appreciated in Eq. 2.

$$S_j = \frac{1}{N}\sum_{i=1}^{N} S_{ij}$$
$$O = Max\{S_1, S_2, ..., S_j\} \tag{2}$$

Here S_j is the predicted probability for the j_{th} class, S_{ij} is the prediction for the j_{th} class of the i_{th} classifier, and O is the final prediction (maximum class average). One problem faced when using these methods is the assumption that all models perform almost equally, which could not be the case.

MBWV is a method based on a metric (frequently accuracy) that weighs each model according to its performance. Those weights can be linear or exponential, depending on the level of penalization wanted for each model. It presents the advantage that no assumption of equality of performance is made, but its estimation is computationally more expensive as it needs an additional metric to be calculated (Eq. 3).

$$\alpha_i = \frac{e^{-f(1-MT_i)}}{\sum_{i=1}^{k} e^{-f(1-MT_i)}}$$
$$MWV_j = \sum_{i=1}^{N} \alpha_i * W_{ij} \tag{3}$$
$$O = Max\{MWV_1, MWV_2, ..., MWV_j\}$$

In this case α_i corresponds to the updated metric or weight (accuracy, recall, precision, etc) of the i_{th} classifier, MT is the selected metric, f is the penalizing factor (f=10 typically) and W_{ij} are the corresponding predictions for the j_{th} class of the i_{th} classifier. Then, MWV_j corresponds to the weighted vote of each model for the j_{th} class and the final decision is selected from the maximum value (Rehman et al. 2021).

MV and AV are different from MBWV because they do not need unequal weights. This could produce, dependent on the data quality and the base model performance, different outcomes. For one side, if the base models have similar performance, AV could be enough for the final prediction, but in the case that the strength on each model is more dissimilar, MBWV could be a better approach (Zhou 2012).

2.3.2 Feature-level fusion ensemble

Stacking and training of a multilayer perceptron (MLP) were the methods used for feature-level fusion. This term refers to fusion before the softmax layers in the CNN, where very abstract features are mixed up. Stacking is the most common meta-learning method. It works as follows: the same data is used to train base learners (or level 0 models), and they all make a prediction. Those predictions (in our case, abstract features) are used by another model (Logistic Regression, Random Forest) to learn how to combine and interpret their results (and produce better predictions) (Mohammed and Kora 2023). It involves two (or more) learning stages and more data: data used for training CNN should not be used as an input to the meta-model, since the base models have been adjusted with this information.

Stacking is adopted in many cases for its flexibility and scalability. That means that if a new base model or architecture is proven to be a good fit, a pipeline could be developed for building an ensemble with that new model incorporated. Its nature also makes it suitable for parallel computation. In contrast with bagging and boosting, this method can achieve good results using just two or three base models, instead of the need of many models for the small variations in data distribution expected to get diversity in bagging (Ting and Witten 1999). Random Forest is a typical bagging example. In case a more complex model is selected (like a CNN), bagging in the more traditional case becomes extremely expensive and hence, unfeasible. Those were the reasons behind the selection of stacking in the feature-level fusion methods.

In comparison, MLP training could be a more suitable way to mix the CNN features than stacking. In intermediate fusion, the features can be concatenated and fed to a simpler neural network. In that way, an enriched mixture of the characteristics detected by each CNN could be taken into account.

The training of that MLP consists of the previous training of the base CNN, the elimination of the classification layer, the concatenation of the feature vectors before the classification layer and the usage of those representations to train a new, smaller network. The benefits in this approach could be the refinement of features beyond each CNN by the MLP instead of assuming equal model importance (voting methods) and the direct integration of the MLP as a head in top of the base learners (in this case, CNN). In comparison with the base CNN, the head is lighter and increases the computational cost slightly. It is a feasible option when using CNN as base models, compared with bagging or boosting.

However, the problem is again model complexity (much higher than late fusion methods) and the quantity of data needed to avoid overfitting. Also, there may exist a saturation in models' predictive power already, and therefore, little or no improvement at all.

3 Results

Metrics considered in our results include accuracy, precision, recall, and f1-score as the four basic measurements. These are calculated from the following concepts:

- True Positive (TP): Number of correctly classified positive (illness) samples
- True Negative (TN): Number of correctly classified negative (healthy) samples
- False Positive (FP): Number of samples incorrectly classified as positive
- False Negative (FP): Number of samples incorrectly classified as negative (missed case)

In this context, accuracy refers to the ratio between the correctly classified samples and the total number of samples, precision gives us the proportion of the correctly classified samples and all samples assigned to that class, recall (or sensitivity) tells us the rate of positive samples correctly classified (it is the capacity to distinguish the disease) and finally f1-score that is the harmonic mean of precision and recall (useful on imbalanced classes) (Hicks et al. 2022). Eqs. 4 to 7 present how each metric is calculated.

Like many of the works consulted, we incorporated the area under the receiver operating characteristics curve (AUC-ROC) in our metrics. In contrast, we also used the Matthews correlation coefficient (MCC) to complement AUC-ROC. This is done because MCC uses TP, TN, FP, and FN in its formula (which can be understood as a correlation coefficient), and it can provide a more reliable estimation of the performance. A high MCC always indicates a high value for each of the four basic rates (and it's robust against imbalanced datasets). In contrast, a high AUC-ROC can guarantee only one high value among sensitivity and specificity in the worst case, and high values for both in the best case. This can lead to inflated or overestimated performance (Chicco and Jurman 2023). In Eq. 8 can be seen the calculation of MCC.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{4}$$

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$F1 - Score = 2 * \frac{precision * recall}{precision + recall} \tag{7}$$

$$MCC = \frac{(TP * TN) - (FP * FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{8}$$

After training our base models for each module, we achieved the results in the internal testing reported in Table 2. Although diverse architectures were used, both modules achieved similar metrics for individual CNN. All metrics go from 0 to 1, except for MCC, that goes from -1 to 1, 0 corresponding to no correlation, -1 to inverted classification and 1 to a perfect classifier. As appreciated, AUC-ROC is more optimistic than all the other metrics and reporting it alone could be misleading. All metrics were considered using macro average to account for class imbalance.

An extension of these results can be seen in Table 3. Divided by module, the different ensemble techniques were applied to each module alone. The intention was to validate the performance enhancement against an individual model and image modality. MBWV refers to the method with no exponential weights where the Eq. 3 is modified to just divide the metric performance of a specific CNN over the sum of all model metrics, and $MBWV_e$ uses Eq. 3 for weight calculations. The metric used for model weighting was precision. For the stacking technique, we extracted the features from the last dense layer before the softmax layer. Then we concatenated them and fed to a Support Vector Machine (SVM). In this two-stage model, we had to separate

Table 2 Internal testing for both modules

Model	Accuracy	Precision	Recall	F1-Score	MCC	AUC-ROC
CFP module						
ResNet50	91.16%	89.09%	91.80%	90.17%	0.80	0.966
DenseNet169	90.89%	88.62%	91.74%	89.83%	0.80	0.966
InceptionV3	89.73%	87.50%	90.21%	88.57%	0.77	0.957
OCT module						
ResNet50	92.93%	92.93%	92.93%	92.93%	0.85	0.9796
Xception	91.98%	92.03%	92.04%	91.98%	0.84	0.9732
MobileNetV3	91.44%	91.49%	91.51%	91.44%	0.83	0.9715

Table 3 Ensemble techniques in internal testing databases for both modules

Technique	Accuracy	Precision	Recall	F1-Score	MCC	AUC-ROC
CFP module						
MV	91.70%	89.46%	92.71%	90.72%	0.82	0.966
AV	91.70%	89.46%	92.71%	90.72%	0.82	0.966
MBWV	91.79%	89.53%	92.86%	90.82%	0.82	0.969
$MBWV_e$	91.88%	89.71%	92.84%	90.93%	0.82	0.969
Stacking (SVM)	91.25%	93.74%	92.40%	93.04%	0.81	0.963
MLP	91.70%	89.25%	93.04%	90.67%	0.82	0.973
OCT module						
MV	94.02%	94.06%	94.05%	94.02%	0.88	0.985
AV	94.02%	94.06%	94.05%	94.02%	0.88	0.985
MBWV	94.02%	94.06%	94.05%	94.02%	0.88	0.985
$MBWV_e$	94.02%	94.06%	94.05%	94.02%	0.88	0.985
Stacking (SVM)	94.02%	92.45%	95.83%	94.05%	0.88	0.978
MLP	93.89%	93.89%	93.89%	93.89%	0.87	0.985

our testing data into subsets for training and evaluation of the SVM. This had to be done to prevent training the SVM with the same data used to train the base models, preventing data leakage and limiting overfitting. Although the SVM did not see the exact same data in the way it was observed by the CNN (SVM only saw a concatenated feature vector) the adjustment with the same training data would be redundant. To select the best hyperparameters for this meta model, we first performed a search in a predefined hyperparameter space. Then, we validated this approach with 5-fold cross-validation and finally, we used all the available data (i.e. the complete testing database) to train the meta-model with the selected parameters. The same extracted features were used for training the small MLP on top of the CNN. For this MLP, we used three dense layers, each with its respective regularization (L2 = 1e-4, batch normalization), with the Adam optimizer. The MLP had sizes for layers of 256, 128 and 64 neurons.

Evaluating a different dataset could help understand the model's real performance. After these results, we used two datasets as external validation. These datasets had pairs of images belonging to glaucomatous or non-glaucomatous aspect, for both OCT and CFP modalities. Our first database is public. It's named "Data on OCT and fundus images for the detection of glaucoma" (Raja et al. 2020). It comprises 50 images, OCT centered in ONH, and a full CFP image. The system for image acquisition was a TOPCON 3D OCT-1000 system, and images were obtained from the Armed Forces Institute of Ophthalmology (AFIO) in Pakistan. They provide a csv file along with the images, where each of the four ophthalmologists provides CDR annotation and their label for the patient (suspect, glaucoma, non-glaucoma). We took these labels and used them for a final labeling, where the final label was assigned to the image, just in case most ophthalmologists agreed. We excluded suspicious cases and obtained 15 glaucomatous cases and 23 control cases to test our approach.

The second database used is a private one. This data was provided by the Instituto Mexicano de Oftalmología I.A.P (IMO) as a collaboration with Universidad Autónoma de Querétaro (UAQ) for research interests. This database consisted of 16 glaucoma cases and 109 control cases. There also existed suspicious cases, but in this first study, we excluded them. In this case, also four ophthalmologists provided their image labelling based on the aspect (image) of each case. Their data was taken with a NIDEK Retina Scan Duo 2 at Querétaro City, México. OCT was also centered in ONH, and the full CFP image was provided.

The preprocessing done to the images used for CNN training was performed for both databases, and the different ensemble techniques were tested. This data was not observed or used by the models in their training phase and is from a different geographical population than the training data. We expected a drop in metrics, because this is a well-reported issue when domain shifting or different data distribution (Diaz-Pinto et al. 2019; Zhou et al. 2023), so we decided at this point to combine both modules to validate the enhancement. The combination of the modalities was chosen to be simple: a simple weighing was done, and the CFP model was selected as the predominant modality due to the data diversity during its training. In Tables 4 and 5 the results of these external validations are presented.

Finally, we provide some examples of our AUC-ROC curves of both internal and external testing for each module trained. Our curves shown in Fig. 7 are from ResNet50 and Xception models for CFP and OCT modality, respectively.

Table 4 Ensemble techniques and individual CNN in the external testing database 1

Technique	Accuracy	Precision	Recall	F1-Score	MCC	AUC-ROC
Data on OCT and fundus images database						
MV	84.21%	84.64%	83.47%	83.81%	0.68	0.881
AV	84.21%	84.64%	83.47%	83.81%	0.68	0.924
MBWV	84.21%	85.80%	84.21%	84.03%	0.69	0.953
$MBWV_e$	**86.84%**	**87.97%**	**86.39%**	**86.61%**	**0.74**	**0.965**
Stacking (SVM)	78.95%	81.45%	80.39%	78.89%	0.61	0.915
MLP	81.58%	84.78%	84.09%	81.57%	0.68	0.924

The highlighted results are the best

Table 5 Ensemble techniques and individual CNN in the external testing database 2

Technique	Accuracy	Precision	Recall	F1-Score	MCC	AUC-ROC
IMO database						
MV	92%	**84.75%**	81.46%	82.98%	0.66	0.853
AV	**93.60%**	83%	**87.03%**	**84.85%**	**0.69**	0.862
MBWV	**93.60%**	83%	**87.03%**	**84.85%**	**0.69**	0.865
$MBWV_e$	92.80%	82.54%	84.39%	83.43%	0.66	**0.871**
Stacking (SVM)	82.40%	81.91%	68.70%	71.64%	0.48	0.846
MLP	89.60%	80.71%	76.59%	78.41%	0.57	**0.871**

The highlighted results are the best

4 Discussion

In Table 2 it can be seen that all trained models converge to a similar local minimum. This could have occurred because the quantity of data for training the models is still not enough to exploit the full potential of a CNN approach. A way to improve these results may be to extend the quantity of available images. We tried to overcome this during training, with our data augmentation pipeline. Even with this approach, during training, we did see moderate overfitting on the training set. Above the 88–90% performance, training and validation curves started to slightly apart. We limited it with our early stopping approach, which stopped the training when the validation loss function stopped improving. More aggressive regularization could mitigate this issue, but in our experience, for this application, it worsened the performance. It could have been for the excessive penalization to the network, producing an inability to learn effectively. We also used different batch sizes. We saw in literature that it is common to use small batch sizes to introduce some noise in the weight updates, allowing to escape local minima. In our case, too small batch sizes caused the models to stop improving, and larger batch sizes caused the learning to be slower. Our final batch was 16 after the experimentation. The fine-tuning approach also could have influenced this. By unfreezing the layers in the base model, a large quantity of parameters becomes available to update suddenly. Unfreezing less layers caused a plateau in the learning curves, but the separation between curves was mitigated. We still maintained the unfreeze because it allowed us to achieve better results, but we point out that the compromise between risk of overfitting and performance on different domains with transfer learning should be evaluated. A way to mitigate this could also be to have more examples during training, along with the augmentation techniques to ensure at least artificially diverse images. This table suggests the performance of each model is similar.

The results in Table 3 show that AV and MV produce almost the same performance, but a slight improvement above the best individual CNN (around 1%). This indicates that all the models make similar mistakes or a peak in learning is generally achieved, so limited complementarity can be expected across them. It can also be seen that $MBWV_e$ or MBWV are enough to combine the models simply yet meaningfully, but without visible enhancement in the ensemble performance. What we saw behind these two techniques, was the fact that even a moderate drop in precision in one model, caused their relative weighting to be penalized severely, limiting its selection during predictions. This penalization was more severe with the based weighted voting exponential version. Moving to the feature fusion-based approaches, stacking and MLP show almost equal performance again, with a slight improvement in some metrics or decay in others. In this internal ensembling by modality, it can be confirmed the model's comparable capacity, because no ensembling technique stands out against the other ones. Counterintuitively it can be appreciated than more complex models (stacking, MLP) do not provide a better combination of the base models, when the expectation was the greater performance of those approximations. However, this could be due to data quantity: these techniques were trained with a second portion of the available data (much smaller

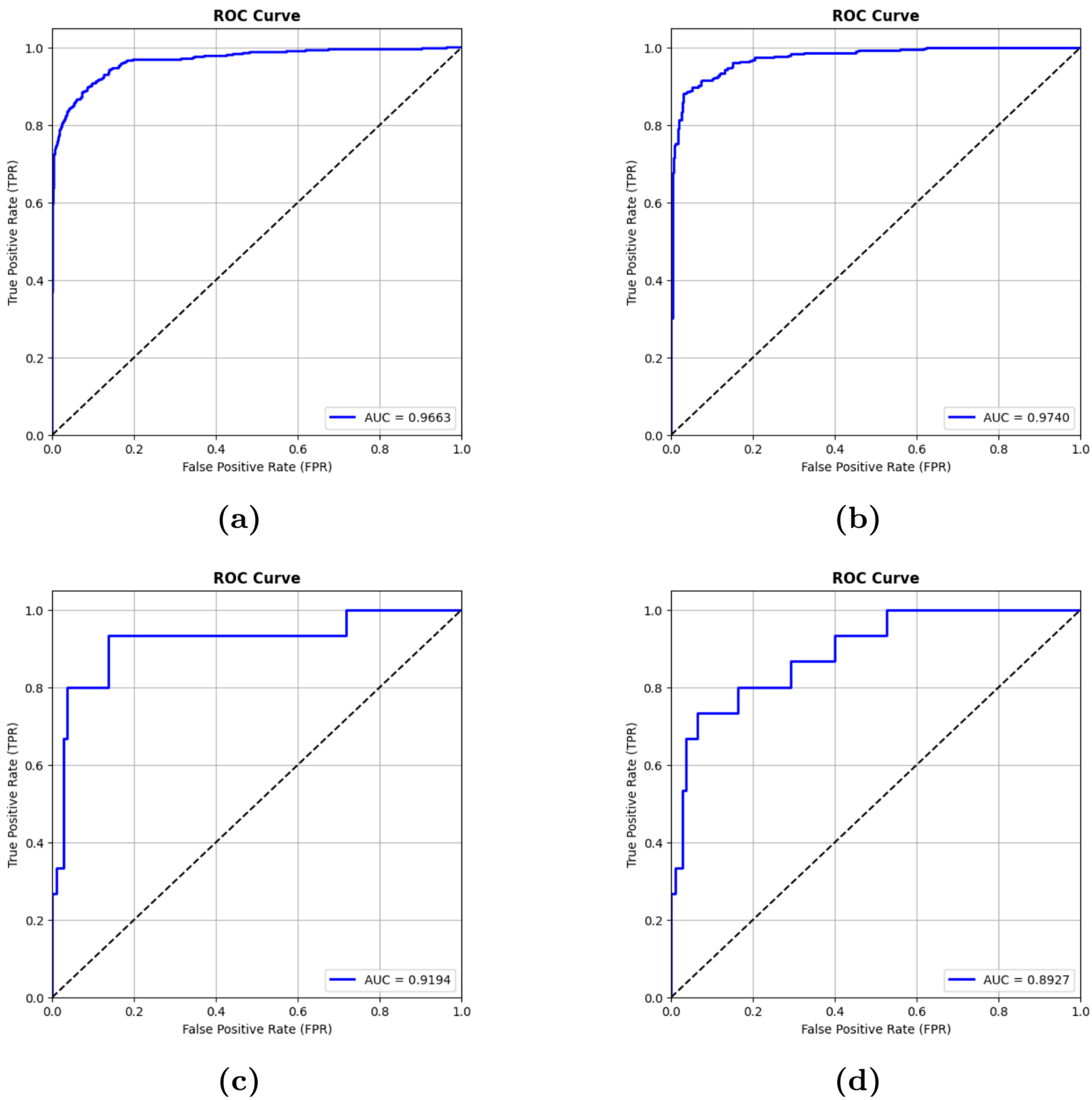


Fig. 7 AUC-ROC curves for **a** CFP internal testing, **b** OCT internal testing, **c** CFP external testing (IMO database) and **d** OCT external testing (IMO database)

than the one used to train CNN). As we mentioned before, for ensuring a fair training of the meta-model, and avoid data leakage, the original testing database was separated into training and testing sets. This allowed us to experiment with the main problem during stacking: it is costly to data. The MLP also used this data partition to be trained, with the intention to choose the best features and model outcomes. Compared with the previous table, these complex ensembles also surpassed the individual trained models. This is expected in an ensemble. Again, the results suggest all the models are similar in their performance. In that case, AV or MV are enough to use as model fusion. This aligns with literature findings, where some authors prefer this combinatorial approach (Velpula and Sharma 2023; Cho et al. 2021).

In external data testing, from Tables 4 and 5, AV produces a good approach despite being simple. This is not necessarily bad, because it requires less memory than calculations done by weighted methods (MBWV) or feature-based ones, which train another model. There exists a decay in metrics in these second methods. This could be due to their complexity. It can be seen in internal testing that the performance is equivalent to or just under simpler methods. We assume that behavior is due to the amount of data available used for training the top model in combination with the good general performance of the base CNN. In fact, stacking and MLP were trained with less than 1000 images for OCT and slightly above that number for CFP images. We also attribute this unexpected drop to the overfitting in both techniques. With that small data quantity to train each meta-learner and the capacity of the base models, the meta-learners could have specialized in the classification on that specific set of examples. On the other hand, the combinatorial approaches provide better results. In the first database, the superior performance is found on the MBWVe approach. In the second one, AV and MBWV achieve the same outcome. This also confirms our results seen on the internal testing, where models perform similarly. That could cause MBWV to learn similar relative weights for each model, transforming practically into AV. This is also seen in the testing database 1, when they achieve almost the same results. MBWVe penalizes models more, so this technique finds relative weights that are more diverse, even with small differences in precision. What these results suggest in both tables is that type of combination does not always produce the best possible result, but it is comparable to the classical fusion methods. The differences in the performance of this fusion method also suggests the base CNN behaves differently for each database, expected for the device acquisition and image source. In both cases (Stacking and MLP) it seems that the complexity combined with this lack of data for the model training causes the system to overfit the training data and then leads to poor generalization. The weighting of the base CNN is the method that, across datasets, provides better balance between metrics and good performance.

In the AUC-ROC curves presented in Fig. 7, it is noticeable that the individual CNN in the CFP module generalizes better than the OCT module despite both models perform well. This behavior could be understood if we remember that we used several databases for CFP module and in comparison, our OCT module relied on just one database. It was also influenced by the amount of data used to train the base models. With more data, less drop is expected on an external database. This causes one module to recognize different variants of optic neuropathy but the other is only focused on POAG cases. Ideally, if we take our external databases with enough cases for both classes and retrain our models, our performance would be significantly better. However, this is not possible as we have only a few examples that present glaucomatous characteristics. Even with this, it can be seen that the distinction of classes by the models on different thresholds is always way above a random guess. Our achieved AUC-ROC values are above 0.85, which is considered high. However, we see plain zones in our model's true positive rate. This is because the presented staircase shaped AUC-ROCs are for an imbalanced database, in which the negative class samples (i.e. healthy eyes) outnumber the positive ones, causing more frequent updates for that class when the thresholds are changed. A more robust comparison should be made with the existence of a larger, balanced database, which may reveal an AUC-ROC curve more representative of the model's performance. This is one of the reasons we included MCC, because it is more conservative and accurate on imbalanced databases.

In terms of explainability, our system belongs to the Deep Learning category. Because of that, it is challenging to explain how CNN determines the presence or absence of optic neuropathy. The well-known Gradient-weighted Class Activation Mapping (Grad-CAM) is the method we use to provide visualization. It can be understood as a heatmap built by the convolutional features chosen by the network, and it shows the areas of the image most crucial for classification. To use it, we extracted the features from the global average pooling layer on CNN for visualization. Figure 8 shows these activations in two of our trained CNNs. The figure shows that for glaucoma detection, the regions highlighted in the two modalities are the ONH or its surroundings. Many other authors have also pointed this out, and it justifies the cropping of the original images (CFP) despite losing some information on other sections of the images. In OCT, it shows that it is possible to estimate if a person shows characteristics related to glaucoma in a B-Scan centered in ONH (visible in the shape of the optic nerve head, area, depth and size of retinal layers, for example).

The mistakes made by the model are also discussed. In Fig. 9, we show some of the model's False Negative (glaucomatous classified or detected as normal aspect) and False Positive (normal aspect detected as glaucomatous) cases. In these pairs of images can be distinguished several characteristics:

1. One of the OCT images (third False Negative) despite having a relatively high OC depth, fails in glaucoma detection. This could be associated with narrowness of the OC what could cause this modality to fail in detect optic neuropathy.
2. In one of these images (False Negatives), there is an elevation of the zone in the ONH, maybe due to papilledema or another defect associated with IOP or other characteristics. This could mean that the model simply observes higher excavations and associates that with glaucoma cases but fails when these large excavations are hidden by anomalous ONH elevations due to inflammation (meaning undetected or masked glaucoma).

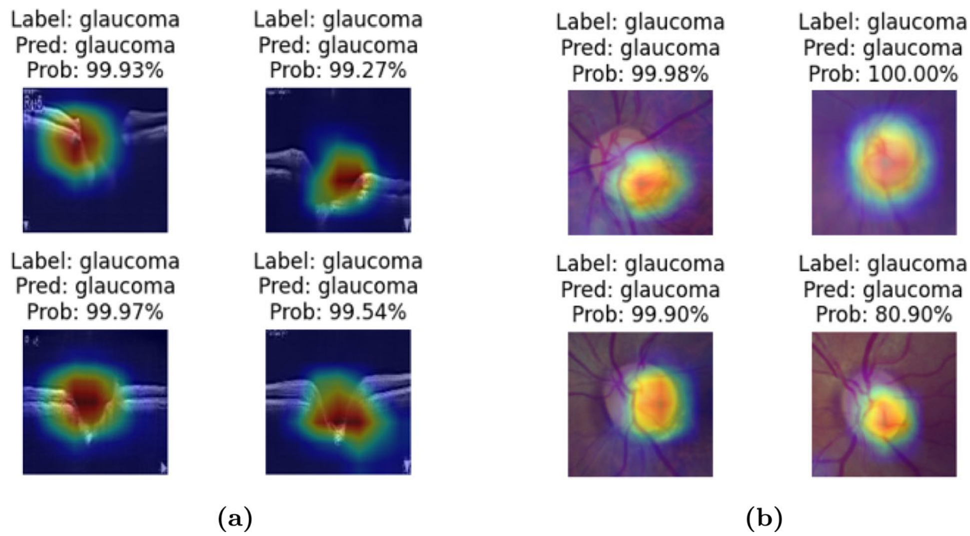


Fig. 8 Grad-CAM visualizations of glaucoma images in **a** OCT, **b** CFP. Images correspond to IMO database

3. In CFP modality, it can be seen that in one image there exist defects that may impede the correct visualization in ONH, and in the others, the size of the optic cup may be the problem that confuses the model.
4. In the false positive cases, it can be noticed that in two cases, OCT and CFP modalities disagree. Our model relies primarily on CFP. Therefore, we biased it to the detection of glaucoma features based primarily on CFP, although our OCT modality in these cases correctly identifies it as normal aspect. This could be avoided in newer versions of our approach, if we collect enough data from different devices for the OCT module, to ensure the model is more robust for this modality.

We selected the ensemble method built with the MBWVe as our final selected approach. This was because it was the technique that in our internal testing achieved slightly better results, and showed superior performance that MV, commonly used for this type of CNN ensembles. Even with the fact that it was not the best performing model in the external testing database 2, we prefer it because it allowed us to use only the better models on each modality and punishing the worse performing ones. In case there exists a later network integration (one or many), this approach limits the impact of a deficient, inferior model.

In comparison with existing ensemble developments, there is the work of Velpula and Sharma (2023) in which five pretrained CNN for glaucoma detection using MV for the fusion was performed. They trained independently on ACRIMA and RIM-ONE databases for their two-class classification. In their reported results ACRIMA database attains higher performance than RIM-ONE. They did not explore other fusion techniques than MV, and the small database size could mask some level of the database memorization. The work developed by Cho et al. (2021) used 56 models (variants in FC layers, preprocessing) for building their ensemble on CFP images. The fusion technique they used was AV and their total amount of images was 3460, acquired with the same device, on two Hospitals with highly population-specific subjects. They excluded images with coexisting optic neuropathies and other retinal pathologies. Their investigation suggested the investigation of other preprocessing techniques along with weighing their individual models for comparison. This study reported accuracy and AUC-ROC of the best single CNN and the ensemble, but the extreme quantity of CNN trained limits its possible application on a real-world environment. The ensemble in Rehman et al. (2021) used five databases for CFP glaucoma classification with a total of 1989 images. They used four CNN as their backbone and used diverse fusion techniques for the ensemble. It is not clear if they trained on each database alone or mixed them up and then validated on a portion of those databases. We assume it was the first approach. If that was the case, external database validation was not performed, and retraining was executed. They also used NasNet-Large as one of their CNN in the ensemble. That model is a heavier one and would have more severe overfitting issues than other ones. For comparison, ResNet50 has almost 3.5 times less parameters than NasNet-Large. Along with these three studies, we present our results and with some of the studies which use CFP, OCT or both modalities (discussed in Sect. 1) on Table 6. On the table there can be seen some of the results on each work. The direct comparison is difficult, because all the works differ in their approach, quantity and diversity of data available. What we can note is that none of these works separated one database for external validation. Instead, all of them retrained on each database separately or mixed databases and trained and tested on examples from the same database (as on our internal approach). Just two of the works consulted used a

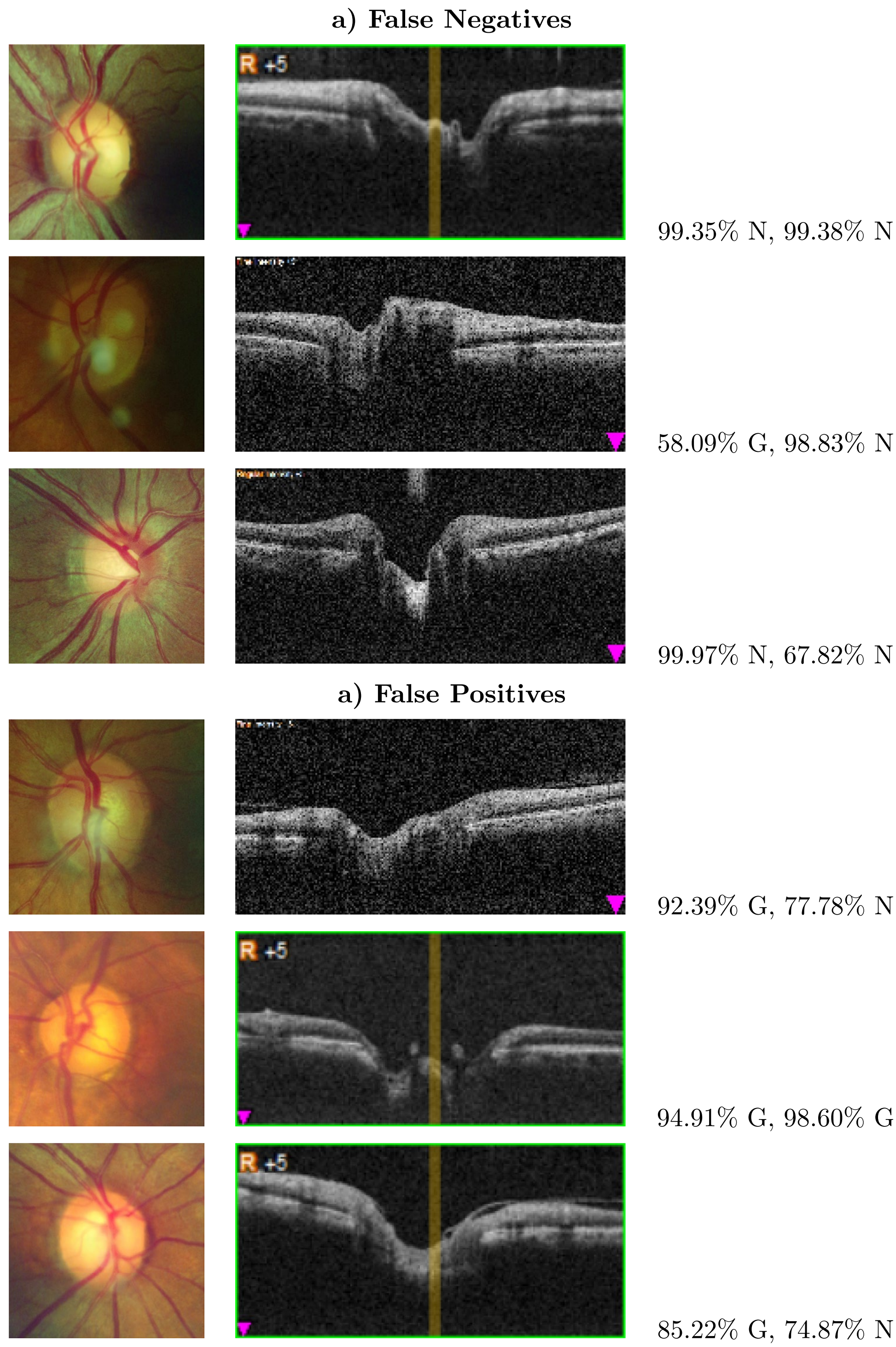


Fig. 9 Incorrect classified cases for IMO database. The probabilities assigned by the model are included (CFP, OCT) G = glaucomatous, N = normal aspect

Table 6 Comparison of proposed model with existing work

Work	Method	acc	prec	rec	F1	auroc	Details
Shehryar et al. (2020)	SVM	96%	-	100%	-	-	118 textural features extracted from CFP. CDR as only feature extracted from OCT. CFP is tested on three databases. OCT tested on the same image source used for training. CFP + OCT approach
Babu et al. (2015)	SVM	96.92%	97.91%	98%	97.95%	-	No external validation. SVM outperformed BPN. ISNT rule and other features calculated. SVM as best classifier and feature engineering processes to extract relevant characteristics. CFP + OCT approach
Saha et al. (2023)	CNN	97.4%	-	97.5%	97.3%	99.3%	Simplified YoloNet for ROI extraction followed by MobileNetSmall classification. Mixture of several databases for training. 10 cross-validation. No hold-out database for external validation. No visualization techniques. CFP approach
Maetschke et al. (2019)	CNN	-	-	-	-	94%	Agnostic approach (No segmented or layer delimitation). Custom 3D-CNN for detection at volume case. AUC reported correspond to test data. No external validation. 1110 scan volumes. CNN outperforms features extracted by the device who took the images used to train diverse ML models
Akter et al. (2022)	CNN	98.6%	97.8%	99.4%	98.59%	99%	Freehand segmentation of the ONH surface area. Images used to train ResNet18, VGG16 and custom CNN. 360 binary segmented images for training. Result in table correspond to custom CNN as it is reported as best performance. All images are from the same device. OCT approach
Shyamalee et al. (2024)	CNN	98.97%	-	99.41%	-	99%	Training and testing for each database separately. Segmentation using modified Attention U-Net (encoder with ResNet50) and subsequent classification using Inception V3. Results in table correspond to the best performing model (RIM-ONE). No external testing. CFP approach
Cho et al. (2021)	CNN ensemble	88.1%	-	-	-	97.5%	56 CNN for the ensemble based on two base architectures. Different filters applied in preprocessing stage (6) along with one or three FC layers. The ensemble mixes the outputs via AV. Images from the same population and device. 3460 images for final database. Results are from the ensemble and 10-cross validation. No external validation. CFP approach
Rehman et al. (2021)	CNN ensemble	99.5%	-	99.1%	-	-	Four CNN ensemble. Explored AV, MV and weighted voting. Different databases used but tested individually. 10 fold-cross validation. 1989 total images. Results are from the ensemble on the best performing combination scheme (weighting) but it is not clear on which database. No external validation. CFP approach
Velpula and Sharma (2023)	CNN ensemble	94.95%	95.40%	94.47%	94.47%	-	Five CNN ensemble. Diverse architectures retrained on each database used. Models combined using MV. Results correspond to RIM-ONE database. 5-cross validation. No external validation. CFP approach
Proposed	CNN ensemble	86.84% 92.80%	87.97% 82.54%	86.39% 84.39%	86.61% 83.43%	96.5% 87.1%	Six CNN ensemble. Explored AV, MV, weighted voting, stacking and MLP. Different databases used for training the CFP module, one to train the OCT module. 7721 CFP images, 4907 OCT images. Results are from the ensemble on both modalities on two external databases. CFP+OCT approach

mixture of modalities, but with problems like the disparity of their modules features (118 vs 1 feature in one approach), or the usage of clinical features no longer available, for the violation in most of the population. Even their total training data was limited in examples. Compared with one modality ensembles, our approach used just three networks by modality, against the four (including NasNetLarge, a resource intensive network), five or 56 used by the other approaches, which makes them more costly if multiple image solutions pretend to be developed.

5 Conclusions

Glaucoma diagnosis and assessment is challenging. It comprehends several variants and numerous studies should be done to confirm or discard their presence or progress. AI has proven to be a promising tool for helping in alleviate the amount of cases examined by specialists. CNN is a cost-effective tool for help in this field, demonstrating a good capacity of distinction between control or optic neuropathy cases focusing on the zones that are typically related to glaucoma, without the need of human-crafted features.

We found that CNN multimodality ensemble is a suitable approach in case model complexity is not an issue, and that in case limited data is available, late fusion methods are the best option for achieving adequate performance (in particular MBWV in both of their versions and AV). Also, multimodality is costly but if done properly, it can be contradictory in certain cases (like our false positives) alerting in difficult cases. About our selected metrics, we encourage other authors to include MCC in their analysis to have a fair and balanced comparison in contrast to AUC-ROC. During our trials, we tried Leaky-ReLU as activation function for dense layers. We also modified the size of those layers. The usage of Leaky-ReLU was done to overcome the problem of dead neurons. We saw an improvement in epoch convergence when using this approach but did not improve our final CNNs. This is why we decided to maintain the original activation function. In the case of more units on each dense layer, the gain in metrics was null or marginal. We also noticed that an effective data augmentation pipeline, with adequate dropout amount, are the most powerful techniques to improve performance. Through the development of this work, we recognize some lessons learned. First, our proposal was a one-step approach. That means classification is performed without any kind of segmentation process. We are strongly convinced that a two-step approach (segmentation + classification) would be more robust and easier to interpret. In fact, Shyamalee et al. (2024) demonstrated their superiority against just classification efforts. The reasons we did not perform it were due to the need for ground truths and the establishment of criterion for custom OC and OD contours by the clinicians that helped this development. We would need an additional effort to perform (or validate the existing ones on the public databases) the contouring. Second, we recognize as valuable the work of Saha et al. (2023) for their findings that efficient designs can yield comparable overall performance than deeper or more parameter networks. This would allow the development of lighter CNN ensemble, allowing for real world applications in resource limited environments. Balancing the quickness and accuracy should be the core of developments. Our next steps would be the development or validation of our approach on lighter networks. Third, a future direction for the enhancement of the results obtained could be the application of Snapshot learning. This could allow us to investigate further, if considering for the final classification different versions of the same model at different timestamps is a potential candidate for an ensemble. If so, the diversity could be achieved by less trained models, and the training time needed would also be shortened. Fourth, an ensemble in the more traditional way (stacking, boosting or bagging) is in general costly to data, either by the overfitting issue present at having many levels of models or the need of using different subsets of the available databases to ensure diversity between model solutions. We do not state that combinatorial or fusion techniques are better than an ensemble, but we recommend that an ensemble for CNN or DL approach should only be addressed in case many images (tens of thousands or more) are available. In case not having this data quantity, we would be inclined to select combinatorial approaches. Fifth, in line with the previous one, a traditional ensemble may be not the only or better option in the case of neural networks. A direct designed term incorporated in the loss function may be a way of an improved performance. Beyond regularization terms, a penalty term that penalizes similar wrong outcomes of different models, and supporting correct decisions of simultaneous trained models, like the proposed by Harangi et al. (2023) could be highly beneficial and an alternative to an ensemble or even combinatorial methods. The incorporation of more clinical factors (like IOP, measurements of studies like gonioscopy, visual field ranges) as a feature into the system could cause an increase in the classification ability. Also, the development of multi-task networks may cause an increased performance in the classification branch, Wang et al. (2020) and could be an interesting approach to validate using multi-modality. Transfer learning and fine tuning the entire networks have also been found to be an excellent starting point. Transfer learning on its own allows an adequate baseline, but without fine tuning, the developed models stagnated. However, the unfreezing of the base model layers increase substantially the risk of overfitting, and aggressive techniques had to be applied to limit this issue. These techniques were dropout, a data augmentation pipeline, and penalty terms. A middle point between unfreezing all the layers and the top ones is recommended to be searched for, as it would limit the overfitting problem and maximize the performance. In our application, the best performance was when we unfreeze all layers. Concerning the OCT preprocessing step, and in-line with MRI advances in denoising, the usage of techniques like NLM-PCA, for example, despite being more costly during preprocessing, may preserve in a better way the textures and edges than the fast NLM options. The usage of different loss functions should also be investigated. There exists a loss function called weighted focal cross entropy loss, which was developed initially to be applied on extremely imbalanced datasets, designed for small object recognition in images with a huge background. It may be applied to the optimization in the classification or detection of OC and OC on both images. The loss is adjusted to be mostly influenced by hard examples, and easier examples contribute fewer. This needs hard examples to be well labeled, high quality and consistent. Otherwise, the model may learn from noise or non-informative examples. A careful review of databases' quality needs to be done before intending to use this loss function as an alternative.

Moving on, our approach still had some limitations. First, our OCT module was trained on a publicly available database (the only one open without registering). This database was of limited resolution, which could have impacted in our model performance, and it was developed to study Primary Open Angle Glaucoma cases against healthier ones. More databases with more resolution, different glaucoma variants and progression, and acquired with different devices, should be used during training the module for an improved generalization. One of our two external databases was imbalanced, and the two of them were small. A more robust result could be obtained from a more balanced database (almost equal cases or at least more proportional) and with more images. MBWV mixes the models consistently between databases. Also, the ensemble techniques used show that, if the base models are powerful enough, and the amount of data is limited, late fusion techniques provide the best results. More research should be done to evaluate the performance of an intermediate fusion technique and its performance against simpler fusion methods. In contrast, the improvement of an ensemble does not surpass 5%. Thus, a trade-off between model complexity (considering quantity of CNN for the ensemble, fusion method) and performance should be the core of developments. The use of lighter models for both segmentation and classification should be further investigated (Saha et al. 2023). Finally, the development of foundation models is one of the most promising approaches found in literature. A foundation model is a large AI model trained on an extensive amount of unlabeled data. That model can be adapted to various use cases, often referred as downstream tasks. One strategy is to use Self Supervised Learning to develop this kind of models. In case of image foundation models, image patches are masked, and the model learns to reconstruct the images (learning specific domain structures). Then with specific labels, those models can be adapted to a wide range of tasks. There exists a foundation model developed for retinal images, so a potential use case for multi-modality, would be the training to a downstream task for both kind of images and apply either a fusion technique or a custom penalizing term to complement modalities. This is a promising approach to tackle multiple disease diagnosis, screening and mitigate the need of many explicit labels. That model demonstrated superior performance to downstream tasks, with limited labeled data (Zhou et al. 2023).

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Author contributions M.B.F proposed the original idea. All authors contributed to refining the study design and framework. Data collection and image labeling of the IMO database were performed by M.B.F. R.G.L. executed data preprocessing, methodology, analysis, selection of CNNs, implementation of methods to build the ensemble, visualization of the saliency maps, and drafted the initial version of the manuscript. G.A.F and S.T.A. validated the methods, reviewed and edited the final version of the manuscript. S.T.A supervised the development of the research. All authors commented on previous versions of the manuscript, read and approved the final manuscript.

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Data availability All data used in this study except the IMO database are publicly available.

Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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